MATH3026/4022 Lecture 1
Time Series: An Introduction

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MATH3026/4022 – Basic Information

Lecturer

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Office hour

Thursdays during term time 13:00–14:00, Office B13, Mathematical Sciences Building (or please e-mail the lecturer to arrange an alternative time).

Moodle

Lecture slides, examples sheets etc. can be found on the MATH3026/4022 Moodle page.

There is also a course discussion forum through which you can post brief questions etc. about the module.

MATH3026/4022 – Basic Information

Timetable:

LECTURES: UoN Weeks 19–27, 32–33

Monday 16:00–18:00, Pope Building, A13

Thursday 09:00–11:00, Pope Building, A14

NOTE: No lecture on Monday 4th May because of the May Day bank holiday.

COMPUTER PRACTICALS: UoN Weeks 22, 23, 25

Wednesday 12:00–13:00: Coates Building, C19

MATH3026/4022 – Assessment

Written Examination: A 3 hour closed book in-person examination to take place during the Spring Semester examination period (May/June 2026).

Same examination for both MATH3026 and MATH4022 students.

All questions will be compulsory (i.e. full marks would be awarded to correct answers to ALL questions on the examination).

The written examination mark contributes 80% of the overall module mark for both MATH3026 and MATH4022.

MATH3026/4022 – Assessment

Coursework for MATH3026: Two take-home coursework assignments, each counts for 10% of the overall module mark for MATH3026.

Assessed work	Date set	Hand-in date
MATH3026 Coursework 1	05/03/2026	19/03/2026, 15:00
MATH3026 Coursework 2	20/03/2026	27/04/2026, 15:00

Coursework for MATH4022: One take-home coursework assignment that counts for 20% of the overall module mark for MATH4022.

Assessed work	Date set	Hand-in date
MATH4022 Coursework	20/03/2026	27/04/2026, 15:00

For both MATH3026 and MATH4022, setting and submission of coursework will take place through the course moodle page.

Mathematical Pre-requisites

A (non-exhaustive) list of mathematical methods that will be used in MATH3026/4022:

- Factorising polynomials and solving quadratic and cubic equations
- Taylor series expansion of a function
- Solution of difference equations
- Complex numbers: equations with complex roots, modulus of a complex number, exponential and modulus-argument forms of complex numbers
- Identifying and summing geometric progressions
- Differentiation and integration
- Some use of trigonometry identities

Statistical Pre-requisites

A (non-exhaustive) list of statistical methods/approaches that will be used/required in MATH3026/4022:

- Expectation and variance – especially for sums of random variables
- Covariance and correlation between random variables
- Probability density functions
- Maximum likelihood estimation
- Hypothesis testing
- Confidence intervals

Lecture 1: Outline

- What are time series?
- Some examples of time series
- Why study time series?
- Objectives when modelling time series in the real world.
- Some key definitions to be used during the course.

In previous statistics courses we have often considered **random samples** of data. For example, we may have seen statements such as:

> *'Suppose that $X_1, \dots, X_n$ are a set of n independent and identically distributed random variables.'*

But what if $X_1, \dots, X_n$ are not independent?

Instead, $X_1, \dots, X_n$ could be **correlated**.

Are the methods that we have used for the analysis of independent data suitable for data that are correlated?

In MATH3026/4022, we shall be concerned with data where **correlation** may occur amongst **a series of variables over time.**

Time Series: Some Examples

For example, suppose we wish to analyse the following datasets:

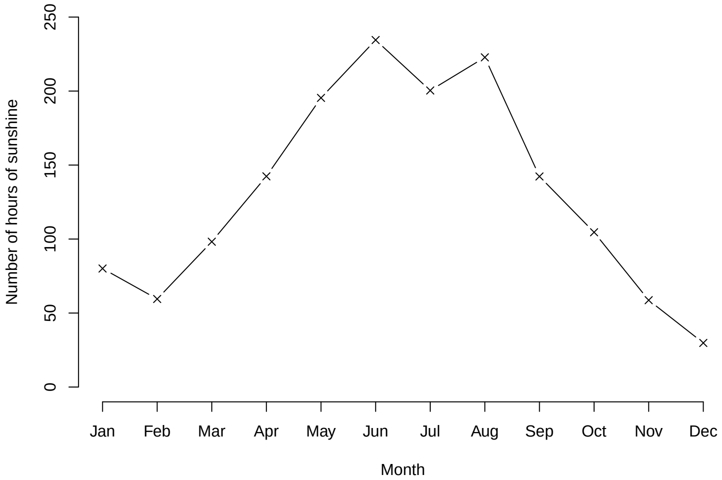
- Number of hours of sunshine each month during a given time period.
- Number of new Covid-19 cases per week during 2020.
- Daily FTSE100 value over a given period.
- Retail sales figures over successive weeks.
- Many more examples...!

We see that datasets like these consist of a **series** of **measurements** made over time.

It is easy to think that **a given data point** in any of these examples **may not** be **independent** of **nearby (in time) observations**.

Time Series: An Example

For example, the plot below shows the number of hours of sunshine for the East Anglia region of England for each month during 2024.



Time Series: An Example

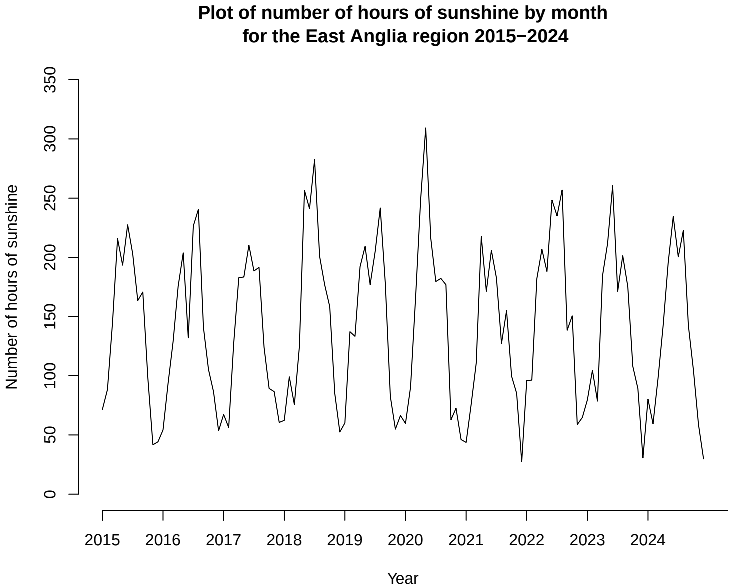
We might expect points that lie close to each other to be more correlated than those that lie further away from one-another.

For example, the number of hours of sunshine observed in May is likely to be more similar to that in April than it is to that in November.

We can also consider similar data across many years.

For example, in the next slide we see a plot of the numbers of hours of sunshine in East Anglia, by month, from 2015 to 2024.

Time Series: An Example



Time Series: An Example

We might notice **seasonal variation** in these data. In addition, measurements in the same month but in different years (e.g. January 2015 and January 2016) may be correlated because they are likely to be similar.

For example, each year, the number of hours of sunshine appears to be low in January before increasing by the middle of the year (and perhaps fluctuating) and then decreasing again by the end of the year.

Whilst the above description - in words - might be of interest, as statisticians we are interested in **modelling** such data.

If we'd like to model data like these then some of the methods with which we are familiar (e.g. linear regression) may not be appropriate. Why?

Why study time series?

A **time series** is simply a collection of **data observations** made **sequentially over time**.

Why are we interested in studying (or modelling) time series?

The **past** behaviour of a variable of interest *may* tell us something about **likely future values** of the variable.

For example, with the hours of sunshine example, knowing how the numbers of hours of sunshine varied during 2020 may allow us to predict what the level of sunshine is likely to be for each month in 2021.

Using the **past to predict future values** is known as **forecasting** and this is something we will cover during MATH3026/4022.

Why study time series?

Clearly, forecasting (being able to 'predict the future') is a popular exercise in many fields. For example:

- **FINANCE**: If we know a company's share price on the UK stock market for the past x months, what are the likely values for the next y months?
- **MEDICINE**: If we know the number of cases of a disease in a particular region for the past x weeks, what are the likely values for the next y weeks?
- **CLIMATE SCIENCE**: If we know the average UK temperatures for the past x years, what are the likely average temperature values for the next y years?
- **RETAIL**: If we know the sales figures for a given store for the past x weeks, what are the likely sales figures for the next y weeks?

Why study time series?

We see that time series data are common across many fields and, clearly, it is of interest to be able to model time series data.

In this course, we are interested specifically in **stochastic** time series models (i.e. those where there is **uncertainty** about future values and uncertainty is **modelled using probability**).

In addition, we shall consider only **discrete time series** where observations are collected at discrete points in time (e.g. every hour, week, month, year etc.). However, continuous time series do exist.

For example, a patient in hospital may have their heart rate or blood pressure measured continuously.

We will not consider continuous time series further.

Why study time series?

Note that when we use the term **discrete time series** we mean that **observations are made** at **discrete time points**. The observations themselves may be continuous variables.

For example, rainfall in millimetres (a continuous variable) measured every day (discrete time points) is an example of a discrete time series.

Our **objectives** in studying or modelling a time series may be classified as **descriptive**, **explanatory**, **predictive** or **controlling**.

We shall now outline each of these in words before proceeding to some mathematical definitions.

Descriptive Objectives

Given a set of time series data, we would usually plot the observations over time.

This is known as a **time plot** and we have seen an example already in this lecture (plot of the Midlands sunshine hours data over time).

A **time plot** would usually be the **first step in any analysis** of time series data and we would then seek to **describe** features of the time series.

A time plot may indicate the presence of a **trend** and/or **seasonal variation**. In addition any unusual observations or **outliers** may also be apparent.

Descriptive Objectives

Seasonality refers to how observations may fluctuate or change in a short, cyclic period.

For example, a company's monthly sales figures may fluctuate within a year.

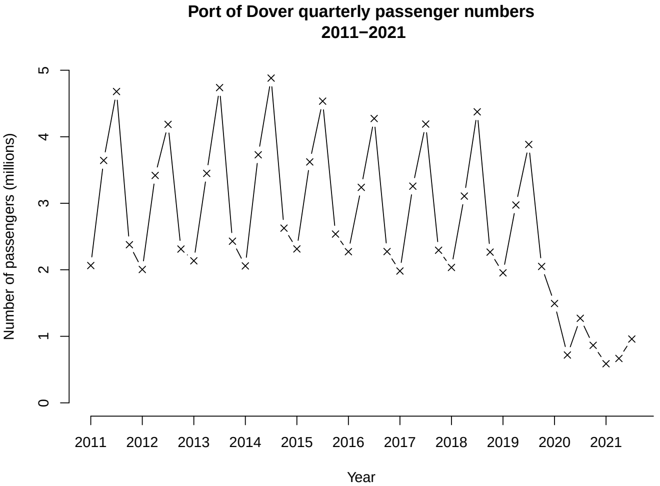
An ice cream seller in a UK seaside resort is likely to have high sales figures in July and August but lower sales figures in January and February. This pattern is likely to repeat each year.

Trend refers to how observations may change in general in the long term.

For example, despite seasonal variation, the ice cream seller's sales figures may increase (or decrease) year-on-year as the business develops or consumer demand changes.

Example 1.1

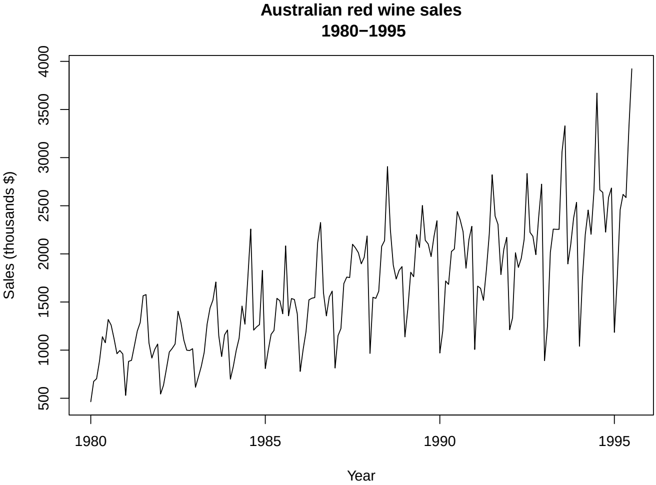
The time plot below shows quarterly passenger numbers for the Port of Dover for the years 2011–2021. Describe the time series in words, considering seasonality, trend and possible outliers.



Example 1.1

Example 1.2

In words, describe the following time series of monthly Australian red wine sales from 1980–1995.



Example 1.2

Explanatory Objectives

Sometimes we can have observations taken over time on two or more variables.

In such situations we may be interested in how a change in one time series affects another time series.

For example, we may be interested in how the weather affects sales of particular goods (e.g. daily temperature and sales of ice creams).

This type of modelling can be performed using an analysis of **linear systems**.

We can use such models in an effort to explain how one time series may be influenced by another (an explanatory objective).

However, we won't cover linear systems any further in MATH3026/4022.

Predictive Objectives

Given an observed time series, we might seek to predict future values of the series.

This is known as **forecasting**.

For example, suppose we have annual sales figures for a business for the last five years.

The business owner might want to estimate sales figures for the next year (or two, three years etc.).

We shall explore forecasting in more detail later on in MATH3026/4022.

Control Objectives

We might seek to control a physical or economic system by analysing a time series.

For example, time series data may be collected on the 'quality' of a manufacturing process and an aim may be to ensure that output data remain within pre-specified quality bounds over time.

Quality control is linked inherently to prediction. It is easy to see that we might wish to predict whether or not a variable of interest will be at an acceptable level (with regard to 'quality') at a future time point and then perhaps take some corrective action if not.

Control objectives are also linked closely to 'decision theory' in that a 'controller' may observe a time series and then take one or more 'actions/decisions' at various points in time to affect future values of the time series.

Key Definitions

Having explored some of the features of time series and why we might model time series data, we shall define some important mathematical terms that will be necessary throughout the course when fitting models for time series data.

Definition 1.1: Stochastic Process

A **stochastic process** $\{X_t \colon t \in \mathcal{T}\}$ is a collection of random variables defined on the same **probability space** $(\Omega, \mathcal{F}, \mathbb{P})$, indexed by some set $\mathcal{T}$ where $\mathcal{T}$ is known as the **index set**.

Often, and especially in MATH3026/4022, the index set $\mathcal{T}$ represents **time**.

Recap: Probability Space

Recall that a **probability space** $(\Omega, \mathcal{F}, \mathbb{P})$ contains three elements that define a random process:

(i) The **sample space** Ω: A non-empty set of all possible outcomes.

(ii) A **set of events** $\mathcal{F}$: A set of all subsets of Ω, (i.e. the set of all possible 'events' that can happen).

(iii) A **probability measure** $\mathbb{P}$: A set function, acting on $\mathcal{F}$ that assigns a probability (i.e. a number in the set $[0, 1]$) to each element of $\mathcal{F}$. That is

$$\mathbb{P}\colon \mathcal{F} \to [0, 1].$$

This assigns a 'probability' to each possible event and the sum of these probabilities is 1.

Key Definitions

If $\mathcal{T}$ is continuous (e.g. $\mathcal{T} = (-\infty, \infty)$) then the stochastic process is said to be **continuous**.

If $\mathcal{T}$ is discrete (e.g. $\mathcal{T} = \{0, \pm 1, \pm 2, \ldots\}$) then the stochastic process is said to be **discrete**.

In MATH3026/4022 we shall consider only discrete stochastic processes and those where members of the index set are equally spaced (e.g. $\mathcal{T} = \{0, 1, 2, 3, \ldots\}$).

Having defined a stochastic process, we shall now define a **time series** as a special case of a stochastic process.

Key Definitions

Definition 1.2: Time Series

A **time series** is a set of **observations** of a **stochastic process** $\{X_t \colon t \in \mathcal{T}\}$ with each observation recorded at a specific time $t \in \mathcal{T}'$, where $\mathcal{T}' \subseteq \mathcal{T}$.

A time series may be written formally as $\{x_t : t \in \mathcal{T}'\}$.

We note that if $\mathcal{T}'$ is a discrete set then the time series is a **discrete time series**.

Alternatively, if $\mathcal{T}'$ is a continuous set the the time series is a **continuous time series**.

We note that the **order in time** is important in the above definition.

Lecture 1: Learning Outcomes

- Understand what is meant by a **time series** and know some common examples of time series.
- Know what is meant by a **time plot** and what is meant by the terms **trend** and **seasonality** when examining a time plot.
- Be familiar with the key objectives of interest when modelling a time series.